

# NITSCI: Cross-Expert Lesion-Aware Attention with Ordinal Dual Heads and Cross-Dataset Transfer for Diabetic Retinopathy Grading

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## Abstract

**Background.** Grading diabetic retinopathy (DR) from colour fundus photographs on the five-level international scale is difficult for deep networks when the target dataset is small: the cues that separate adjacent grades are a few small lesions, the grades are ordered, and the classes are imbalanced.

**Methods.** We present NITSCI, a grading network that couples an EfficientNetV2-S backbone with two token-level experts: a global expert (a Transformer encoder layer over all spatial tokens) and a lesion expert (the same tokens re-weighted by a learned, sparsity-regularised lesion-attention map). The experts exchange information through bidirectional cross-attention and are merged by an image-adaptive gate. A dual head predicts class probabilities and a continuous ordinal grade, which are combined into one score. The model is pretrained on APTOS-2019 ([**napptos**] images) and fine-tuned on IDRiD with five-fold cross-validation; decision thresholds are fitted on out-of-fold predictions only, and the five fold models are ensembled with flip test-time augmentation. All results are reported on a stratified held-out IDRiD test set ([**ntest**] images) that is never used for training, model selection or threshold fitting.

**Results.** On five-class grading NITSCI reaches an accuracy of [**acc5**] % (95% CI [**acc5ci**]) and a quadratic weighted kappa of [**qwk5**] (95% CI [**qwk5ci**]); [**within1**] % of test images are graded within one level of the reference. For referable DR (grade  $\geq 2$ ) it obtains an AUC of [**refauc**], sensitivity [**refsens**] % and specificity [**refspec**] %. Ablations quantify the contribution of APTOS pretraining, the lesion expert, cross-attention, the gate, the ordinal head and the inference-time ensemble. The model has 21.25 M parameters and needs 23.0 GFLOPs per  $448 \times 448$  image.

**Conclusions.** *[AUTHOR: One sentence stating the main finding once results are filled, e.g. which components matter most, followed by: "External validation on independent cohorts is required before any clinical use."]*

**Keywords:** diabetic retinopathy; fundus image grading; lesion-aware attention; cross-attention; mixture of experts; ordinal regression; transfer learning; IDRiD

## Highlights

- A global expert and a lesion expert exchange information through bidirectional cross-attention.
- An image-adaptive gate weighs global context against lesion evidence for each image.
- Classification and ordinal regression heads are fused into one continuous grade score.
- Thresholds are fitted on out-of-fold data; the test set is never used for any choice.
- Bootstrap confidence intervals and paired McNemar tests accompany every comparison.